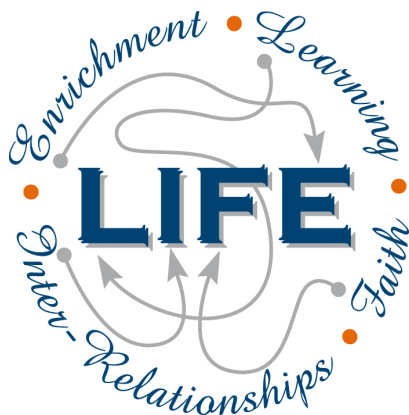




Holy Cross College

Semester 1 Examination 2018

Multiple Choice Booklet

Year 10 Science

Please place your student identification label in this box

Student Name

Student's Teacher

Time allowed for this paper

Reading time before commencing work: 10 minutes
Working time for paper: 110 minutes

Materials required/recommended for this paper

To be provided by the supervisor

This Question/Answer Booklet
Data Sheet

To be provided by the candidate

Standard items: pens, pencils, eraser, correction fluid, ruler, highlighters.
Special items: non-programmable calculators satisfying the conditions set by the School Curriculum and Standards Authority for this course.

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

SECTION 1 MULTIPLE CHOICE (56 marks)

Make a choice of answer and mark it with a cross on the answer sheet in the answer booklet.

1. An absolutely pure substance that cannot be broken down into any other substance is:
 - (a) a compound.
 - (b) an atom.
 - (c) a molecule.
 - (d) an element.

2. The formation of an *ionic compound* from a reaction between the atoms of two elements involves:
 - (a) sharing of pairs of electrons between atoms.
 - (b) donation of valence electrons to the entire crystal structure.
 - (c) transfer of electrons between atoms.
 - (d) ionisation of the atoms of some elements.

3. The *valence electrons* are:
 - (a) the number of total electrons an atom has.
 - (b) the number of electrons in the atom's outer shell.
 - (c) the number of electrons an atom has after it reacts.
 - (d) the number of electrons an atom has before it reacts.

4. Which of the following is likely to be *ionically bonded*?
 - (a) CO
 - (b) Na₂SO₄
 - (c) NO₂
 - (d) F₂

5. The *mass number* of an atom is the total:
 - (a) number of protons and neutrons in the atom.
 - (b) number of protons in the atom.
 - (c) mass of the atom expressed in grams.
 - (d) charge on the nucleus of the atom.

6. The equation shown represents a stage in the formation of rust. What are the values of w, x, y and z (in order) needed to balance this equation?



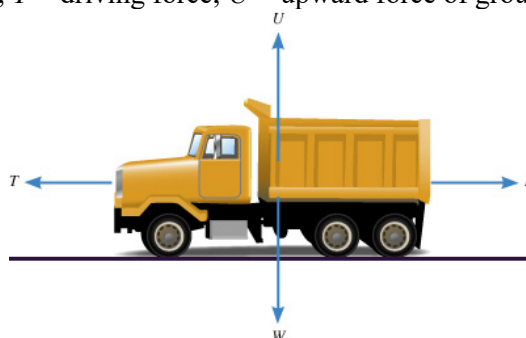
- (a) 2, 1, 2, 2.
- (b) 2, 1, 1, 2.
- (c) 1, 2, 1, 2.
- (d) 4, 1, 2, 2.

7. The atom represented by the symbol $^{52}_{24}\text{Cr}$ has:
- (a) 24 protons, 52 neutrons and 24 electrons.
 - (b) 52 protons, 52 neutrons and 52 electrons.
 - (c) 24 protons, 28 neutrons and 24 electrons.
 - (d) 28 protons, 24 neutrons and 28 electrons.
8. The electronic configuration of the element with atomic number 16 is:
- (a) 2, 8, 6.
 - (b) 8, 8.
 - (c) 2, 2, 8, 4.
 - (d) 2, 14.
9. An atom with the electronic configuration 2, 8, 8, 2 would be expected to form an ion of charge:
- (a) -2.
 - (b) +2.
 - (c) +1.
 - (d) -6.
10. The element chlorine is an example of:
- (a) an alkali metal.
 - (b) an alkaline earth metal.
 - (c) a halogen.
 - (d) a noble gas.
11. The calcium ion carries a charge of +2, the nitrate ion a charge of -1. The correct formula for calcium nitrate is:
- (a) CaNO_{32} .
 - (b) $\text{Ca}_2(\text{NO}_3)$.
 - (c) CaNO_3 .
 - (d) $\text{Ca}(\text{NO}_3)_2$.
12. The formula for dinitrogen tetrahydride is:
- (a) N_2H_5 .
 - (b) N_4H_2 .
 - (c) N_2H_4 .
 - (d) N_5H_2 .
13. If an electron is removed from an atom, which of the following would be *true*?
- (a) The atom becomes positively charged because it now has more protons than electrons.
 - (b) The atom is now negatively charged because it has fewer electrons than protons.
 - (c) The atom is now positively charged because there are more electrons than protons.
 - (d) The atom is now negatively charged because it has fewer electrons.

14. Which of the following compounds is made up of 4 different elements and contains a total of 15 atoms?
- (a) $\text{C}_4\text{H}_9\text{OH}$
 - (b) H_2SO_4
 - (c) $\text{Ca}(\text{HCO}_3)_2$
 - (d) $(\text{NH}_4)_2\text{SO}_4$
15. If an atom with 13 protons lost 3 electrons in a reaction, how many electrons would the particle now have and what would be the charge on the ion?
- (a) 16 electrons; + 3 charge.
 - (b) 10 electrons; + 3 charge.
 - (c) 16 electrons; - 3 charge.
 - (d) 10 electrons; - 3 charge.
16. Use your Periodic Table to decide which of the following group of elements is most alike.
- (a) S, F, He.
 - (b) O, F, Ne.
 - (c) O, Cl, Kr.
 - (d) F, Cl, Br.
17. The number of electrons in a beryllium atom is:
- (a) 4.
 - (b) 9.
 - (c) 13.
 - (d) 5.
18. Which of the following statements about an atom is *correct*?
- (a) The nucleus consists of protons and neutrons.
 - (b) Most of the mass of an atom is due to the electrons.
 - (c) Protons and electrons are found in the nucleus.
 - (d) The electron cloud is positively charged.
19. What is the electronic configuration of magnesium?
- (a) 2, 8, 2
 - (b) 2, 4, 4, 2
 - (c) 4, 2, 2, 4
 - (d) 4, 8
20. In the periodic table, elements with similar properties can be found in:
- (a) every second row.
 - (b) every fourth column.
 - (c) the same group.
 - (d) the same period.
21. Noble (inert) gases are all:
- (a) unreactive.
 - (b) conductive.
 - (c) radioactive.
 - (d) rare.

22. Covalent bonds between atoms are formed by:
- the sharing of electrons.
 - the sharing of protons.
 - the sharing of neutrons.
 - the attraction of ions.
23. The best definition of an ion is:
- an atom that has lost electrons.
 - a type of bond.
 - an atom with too many electrons.
 - a charged atom.
24. Which one of the following equations is **not** balanced correctly?
- $3 \text{ NO}_{(g)} + 2 \text{ O}_{2(g)} \rightarrow 3 \text{ NO}_{2(g)}$
 - $2 \text{ C}_2\text{H}_{6(g)} + 7 \text{ O}_{2(g)} \rightarrow 4 \text{ CO}_{2(g)} + 6 \text{ H}_2\text{O}_{(g)}$
 - $\text{HCl}_{(aq)} + \text{NaOH}_{(aq)} \rightarrow \text{H}_2\text{O}_{(l)} + \text{NaCl}_{(aq)}$
 - $2 \text{ H}_2\text{S}_{(g)} + 3 \text{ O}_{2(g)} \rightarrow 2 \text{ H}_2\text{O}_{(g)} + 2 \text{ SO}_{2(g)}$
25. Given that the ammonium ion is NH_4^{1+} and the carbonate ion is CO_3^{2-} , the correct formula for ammonium carbonate is:
- $(\text{NH}_4)_2\text{CO}_3$.
 - NH_4CO_3 .
 - $\text{NH}_4(\text{CO}_3)_2$.
 - $\text{NH}_4^{1+}\text{CO}_3^{2-}$.
26. Cobalt sulphate is an example of:
- an ionic compound.
 - a covalent compound.
 - a molecule.
 - None of the above.
27. Which element is represented by X in $^{11}_5\text{X}$?
- beryllium
 - boron
 - helium
 - lithium
28. The periodic table:
- is a systematic chart listing all known elements.
 - arranges elements from lowest to highest atomic number.
 - separates the metals and non-metals.
 - All of the above.
29. Positive ions are formed when atoms:
- gain protons.
 - gain electrons.
 - lose protons.
 - lose electrons.

30. Calculate the velocity of a bus that travels between two traffic lights 145 metres apart in 20.0 seconds.
- 7.00 m/s
 - 7.25 m/s
 - 6.25 m/s
 - 7.50 m/s
31. Acceleration is defined as:
- the rate of change of speed.
 - the rate of change of displacement.
 - the rate of change of distance.
 - the rate of change of velocity.
32. A decreasing rate of change of velocity could be called:
- positive acceleration.
 - negative velocity.
 - deceleration.
 - braking.
33. In the absence of air resistance:
- heavier objects would accelerate toward the ground faster than lighter ones.
 - more streamlined objects would accelerate toward the ground faster than less streamlined ones.
 - all objects regardless of mass would accelerate toward the ground with the same velocity.
 - objects with more mass would accelerate toward the ground faster than objects with less mass.
34. What is the acceleration of a car if it increases its velocity from 5.00 m/s to 15.0 m/s in 4.00 s?
- 2.50 m/s²
 - 2.50 m/s
 - 150 m/s²
 - 5.20 m/s²
35. The diagram below shows the four forces acting on a van travelling along a straight, flat road. The forces are:
 R = total resistance force; T = driving force; U = upward force of ground on van; W = weight of van.



At one time, the van is speeding up in the forward direction. Which of the following best describes the forces at this time?

- $U = W$
- $U = W$; $T < R$
- $U = W$; $T = R$
- $U = W$; $T > R$

36. A balloon is inflated. When released, it rapidly speeds up. Which of the following best describes the force that causes the balloon's motion?
- (a) The push of the air in the room on the escaping air.
 - (b) The push of the escaping air on the air in the room.
 - (c) The push of the balloon on the escaping air.
 - (d) The push of the escaping air on the balloon.
37. You are watching a video of a passenger standing in a tram. Suddenly the passenger topples backwards. Which of the following best describes the motion of the tram at this moment?
- (a) The tram starts moving forward.
 - (b) The tram starts moving backward.
 - (c) The tram is moving forwards and it starts to slow down.
 - (d) The tram is moving backwards and it starts to speed up.
38. Which of the following is **NOT** a true statement about your inertia?
- (a) It tends to keep you stationary when a force acts on you.
 - (b) It is a force that keeps you moving at a constant speed.
 - (c) It tends to keep you moving in a straight line.
 - (d) It tends to resist any acceleration.
39. An object accelerates at 2.5 m/s^2 under a net force of 20 N. What is the mass of the object?
- (a) 0.25 kg
 - (b) 6.0 kg
 - (c) 8.0 kg
 - (d) 50 kg
40. A basketball player crouches down, and then stretches his legs to spring into the air. Which of the following best describes the size of the force of the ground on his feet while he is rising from the crouched position? (*Note - he has not left the ground.*)
- (a) Greater than the force of his feet on the ground.
 - (b) Equal to the force of his feet on the ground.
 - (c) Equal to the weight of the player.
 - (d) Zero.
41. How long would a car travelling at a speed of 60 km/h take to cover a distance of 90 km?
- (a) $1\frac{1}{2}$ hours
 - (b) $2\frac{1}{2}$ hours
 - (c) $3\frac{1}{2}$ hours
 - (d) $4\frac{1}{2}$ hours
42. Which of the following is **NOT** a benefit of the 'crumple zones' of a vehicle during a crash?
- (a) They increase the time taken to stop.
 - (b) They spread much of the energy of impact.
 - (c) They absorb much of the energy of impact.
 - (d) They reduce the risk of injury to any pedestrian struck by the vehicle.

43. An object starts from rest and accelerates at 5.0 m/s^2 . After 4.0 seconds, its speed will be:
- (a) 0 m/s.
 - (b) 10 m/s.
 - (c) 20 m/s.
 - (d) 30 m/s.
44. The tendency of an object to resist changes in its motion is called:
- (a) inertia.
 - (b) weight.
 - (c) friction.
 - (d) work.
45. Newton's Law, known as the Law of Inertia, is:
- (a) his first law.
 - (b) his second law.
 - (c) his third law.
 - (d) none of the above.
46. "If an object has zero acceleration, the object must be at rest". This statement is:
- (a) always correct.
 - (b) sometimes correct.
 - (c) never correct.
47. An object is acted upon by a push force of 50 N forwards and a total frictional force of 40 N. The nett force will be:
- (a) 10 N backwards
 - (b) 10 N forwards
 - (c) 90 N backwards
 - (d) 90 N forwards
48. The size of the acceleration acting on an object of mass 10.0 kg that has a nett force of 3.0 N acting on it will be:
- (a) 13.0 m/s^2 .
 - (b) 3.3 m/s^2 .
 - (c) 0.3 m/s^2 .
 - (d) 30.0 m/s^2 .
49. A cyclist has a weight of 500 N, the road pushes up with 500 N, the push force forwards is 30 N and the friction opposing the motion is 32 N. The cyclist will be:
- (a) stationary.
 - (b) accelerating.
 - (c) decelerating.
 - (d) travelling at constant velocity.

50. When you hit a tennis ball with a tennis racquet, the action force is the force of the racquet hitting the ball and the reaction force is:
- (a) the ball hitting the racquet.
 - (b) the jarring force you feel through your hand.
 - (c) the weight force on the ball.
 - (d) the weight force on you.
51. Which of the following is **NOT** a significant benefit of air bags in a collision?
- (a) They prevent collisions between occupants and the windscreen.
 - (b) They prevent collisions between the driver and the steering wheel.
 - (c) They protect occupants from whiplash injury in rear-end collisions.
 - (d) They cushion impacts between the occupant and the side of the vehicle.
52. Which of the following does **NOT** describe a benefit of using a seatbelt in a head-on crash?
- (a) It stops the occupant from continuing to move forward when the vehicle stops.
 - (b) It reduces the risk of whiplash type injury.
 - (c) It stops the occupant at the same rate as the vehicle.
 - (d) It prevents the occupant from striking the windshield.
53. Find the final velocity of a cyclist travelling at 3.00 m/s if she accelerates at 1.20 m/s² for 5.00 seconds.
- (a) 9.00 m/s
 - (b) 19.0 m/s
 - (c) 99.0 m/s
 - (d) 0.900 m/s
54. An object has a mass of 250 kg . On Earth, it would have a weight closest to:
- (a) 2.5 N.
 - (b) 25 N.
 - (c) 250 N.
 - (d) 2500 N.
55. A driver noticed that his speedo reading was 118.8 km/h. Which of the following represents the speedo reading in m/s?
- (a) 11880 m/s
 - (b) 33 m/s
 - (c) 118 m/s
 - (d) 330 m/s

END OF SECTION 1